FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: faultsweep
Model status: BOUNDED_DOCUMENTED_FAULT_LOCATION_SWEEP
Captured UTC: 2026-10-26T06:04:34.981Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: nominalLLV | Entered value: 13800
Input path: voltageFactor | Entered value: 1.1
Input path: frequencyHz | Entered value: 60
Input path: faultType | Entered value: AG
Input path: faultBus | Entered value: F
Input path: buses[0] | Entered value: F
Input path: buses[1] | Entered value: A
Input path: buses[2] | Entered value: B
Input path: branches[0].id | Entered value: line-A
Input path: branches[0].from | Entered value: A
Input path: branches[0].to | Entered value: F
Input path: branches[0].positiveSequence.rOhm | Entered value: 0.06
Input path: branches[0].positiveSequence.xOhm | Entered value: 0.12
Input path: branches[0].negativeSequence.rOhm | Entered value: 0.07
Input path: branches[0].negativeSequence.xOhm | Entered value: 0.14
Input path: branches[0].zeroSequence.rOhm | Entered value: 0.23
Input path: branches[0].zeroSequence.xOhm | Entered value: 0.44
Input path: branches[1].id | Entered value: line-B
Input path: branches[1].from | Entered value: B
Input path: branches[1].to | Entered value: F
Input path: branches[1].positiveSequence.rOhm | Entered value: 0.04
Input path: branches[1].positiveSequence.xOhm | Entered value: 0.13
Input path: branches[1].negativeSequence.rOhm | Entered value: 0.05
Input path: branches[1].negativeSequence.xOhm | Entered value: 0.16
Input path: branches[1].zeroSequence.rOhm | Entered value: 0.22
Input path: branches[1].zeroSequence.xOhm | Entered value: 0.48
Input path: sources[0].id | Entered value: grid-A
Input path: sources[0].bus | Entered value: A
Input path: sources[0].positiveSequence.rOhm | Entered value: 0.08
Input path: sources[0].positiveSequence.xOhm | Entered value: 0.42
Input path: sources[0].negativeSequence.rOhm | Entered value: 0.09
Input path: sources[0].negativeSequence.xOhm | Entered value: 0.46
Input path: sources[0].zeroSequence.rOhm | Entered value: 0.31
Input path: sources[0].zeroSequence.xOhm | Entered value: 1.2
Input path: sources[0].angleDeg | Entered value: 0
Input path: sources[0].emScale | Entered value: 1
Input path: sources[1].id | Entered value: source-B
Input path: sources[1].bus | Entered value: B
Input path: sources[1].positiveSequence.rOhm | Entered value: 0.11
Input path: sources[1].positiveSequence.xOhm | Entered value: 0.6
Input path: sources[1].negativeSequence.rOhm | Entered value: 0.13
Input path: sources[1].negativeSequence.xOhm | Entered value: 0.67
Input path: sources[1].zeroSequence.rOhm | Entered value: 0.33
Input path: sources[1].zeroSequence.xOhm | Entered value: 1.34

Input path: sources[1].angleDeg | Entered value: 0
Input path: sources[1].emScale | Entered value: 1
Input path: faultBuses[0] | Entered value: F
Input path: faultBuses[1] | Entered value: A
Input path: faultBuses[2] | Entered value: B
Input path: faultTypes[0] | Entered value: 3PH
Input path: faultTypes[1] | Entered value: AG
Input path: faultTypes[2] | Entered value: BC
Input path: faultTypes[3] | Entered value: BCG

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: BOUNDED_DOCUMENTED_FAULT_LOCATION_SWEEP
Calculated output path: scope | Value: Fault-location and fault-type grid over one explicitly entered fixed source operating condition. NOT automatic IEC 60909 maximum/minimum source conditions, nor breaker duty or protection certification.
Calculated output path: nominalLLV | Value: 13800
Calculated output path: frequencyHz | Value: 60
Calculated output path: voltageFactor | Value: 1.1
Calculated output path: locations[0] | Value: F
Calculated output path: locations[1] | Value: A
Calculated output path: locations[2] | Value: B
Calculated output path: types[0] | Value: 3PH
Calculated output path: types[1] | Value: AG
Calculated output path: types[2] | Value: BC
Calculated output path: types[3] | Value: BCG
Calculated output path: caseCount | Value: 12
Calculated output path: globalMaximum.faultBus | Value: A
Calculated output path: globalMaximum.faultType | Value: 3PH
Calculated output path: globalMaximum.maximumPhaseRmsA | Value: 30498.57599266589
Calculated output path: maximaByFaultType[0].faultType | Value: 3PH
Calculated output path: maximaByFaultType[0].maximum.faultBus | Value: A
Calculated output path: maximaByFaultType[0].maximum.currentA | Value: 30498.57599266589
Calculated output path: maximaByFaultType[0].minimum.faultBus | Value: B
Calculated output path: maximaByFaultType[0].minimum.currentA | Value: 26978.282592342242
Calculated output path: maximaByFaultType[1].faultType | Value: AG
Calculated output path: maximaByFaultType[1].maximum.faultBus | Value: A
Calculated output path: maximaByFaultType[1].maximum.currentA | Value: 18480.083814940346
Calculated output path: maximaByFaultType[1].minimum.faultBus | Value: F
Calculated output path: maximaByFaultType[1].minimum.currentA | Value: 16620.557944533848
Calculated output path: maximaByFaultType[2].faultType | Value: BC
Calculated output path: maximaByFaultType[2].maximum.faultBus | Value: A
Calculated output path: maximaByFaultType[2].maximum.currentA | Value: 25016.887310037237
Calculated output path: maximaByFaultType[2].minimum.faultBus | Value: B
Calculated output path: maximaByFaultType[2].minimum.currentA | Value: 21969.261655778915
Calculated output path: maximaByFaultType[3].faultType | Value: BCG
Calculated output path: maximaByFaultType[3].maximum.faultBus | Value: A
Calculated output path: maximaByFaultType[3].maximum.currentA | Value: 26616.87717648401
Calculated output path: maximaByFaultType[3].minimum.faultBus | Value: B
Calculated output path: maximaByFaultType[3].minimum.currentA | Value: 23492.242200550376
Calculated output path: cases[0].faultBus | Value: F
Calculated output path: cases[0].faultType | Value: 3PH
Calculated output path: cases[0].maximumPhaseRmsA | Value: 27461.82109261666
Calculated output path: cases[0].groundResidualA | Value: 0
Calculated output path: cases[0].phaseFaultCurrentA[0].real | Value: 6309.743758583883
Calculated output path: cases[0].phaseFaultCurrentA[0].imag | Value: -26727.11640682356
Calculated output path: cases[0].phaseFaultCurrentA[0].magnitude | Value: 27461.82109261666
Calculated output path: cases[0].phaseFaultCurrentA[0].angleDegrees | Value: -76.7168095145296
Calculated output path: cases[0].phaseFaultCurrentA[1].real | Value: -26301.23365750501
Calculated output path: cases[0].phaseFaultCurrentA[1].imag | Value: 7899.159817107831

Calculated output path: cases[0].phaseFaultCurrentA[1].magnitude | Value: 27461.821092616658
Calculated output path: cases[0].phaseFaultCurrentA[1].angleDegrees | Value: 163.2831904854704
Calculated output path: cases[0].phaseFaultCurrentA[2].real | Value: 19991.489898921125
Calculated output path: cases[0].phaseFaultCurrentA[2].imag | Value: 18827.95658971573
Calculated output path: cases[0].phaseFaultCurrentA[2].magnitude | Value: 27461.82109261666
Calculated output path: cases[0].phaseFaultCurrentA[2].angleDegrees | Value: 43.2831904854704
Calculated output path: cases[0].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[0].sourceContributions[0].bus | Value: A
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[0].real | Value:
3942.7531879234994
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[0].imag | Value:
-15207.762296276356
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
15710.548582431029
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
-75.46554491945989
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[1].real | Value:
-15141.685077252243
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[1].imag | Value:
4189.356726544347
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
15710.54858243103
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
164.53445508054014
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[2].real | Value:
11198.931889328744
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[2].imag | Value:
11018.40556973201
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
15710.54858243103
Calculated output path: cases[0].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
44.53445508054013
Calculated output path: cases[0].sourceContributions[1].id | Value: source-B
Calculated output path: cases[0].sourceContributions[1].bus | Value: B
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[0].real | Value:
2366.990570660385
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[0].imag | Value:
-11519.354110547207
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
11760.023957704168
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
-78.3885135761115
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[1].real | Value:
-11159.548580252771
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[1].imag | Value:
3709.8030905634846
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
11760.023957704168
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
161.6114864238885
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[2].real | Value:
8792.558009592385
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[2].imag | Value:
7809.551019983723
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
11760.023957704168
Calculated output path: cases[0].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:
41.61148642388849
Calculated output path: cases[0].kclRelativeResidual | Value: 1.365511307026037e-16

Calculated output path: cases[0].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[1].faultBus | Value: F
Calculated output path: cases[1].faultType | Value: AG
Calculated output path: cases[1].maximumPhaseRmsA | Value: 16620.557944533848
Calculated output path: cases[1].groundResidualA | Value: 16620.557944533844
Calculated output path: cases[1].phaseFaultCurrentA[0].real | Value: 4536.739777344576
Calculated output path: cases[1].phaseFaultCurrentA[0].imag | Value: -15989.400819926514
Calculated output path: cases[1].phaseFaultCurrentA[0].magnitude | Value: 16620.557944533848
Calculated output path: cases[1].phaseFaultCurrentA[0].angleDegrees | Value: -74.15954736119843
Calculated output path: cases[1].phaseFaultCurrentA[1].real | Value: 4.547473508864641e-13
Calculated output path: cases[1].phaseFaultCurrentA[1].imag | Value: 0
Calculated output path: cases[1].phaseFaultCurrentA[1].magnitude | Value: 4.547473508864641e-13
Calculated output path: cases[1].phaseFaultCurrentA[1].angleDegrees | Value: 0
Calculated output path: cases[1].phaseFaultCurrentA[2].real | Value: 0
Calculated output path: cases[1].phaseFaultCurrentA[2].imag | Value: 0
Calculated output path: cases[1].phaseFaultCurrentA[2].magnitude | Value: 0
Calculated output path: cases[1].phaseFaultCurrentA[2].angleDegrees | Value: 0
Calculated output path: cases[1].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[1].sourceContributions[0].bus | Value: A
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[0].real | Value:
2691.562397669685
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[0].imag | Value:
-8875.047550729882
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
9274.210325858794
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
-73.12886548949481
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[1].real | Value:
-80.07152598509356
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[1].imag | Value:
264.44816588912727
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
276.304689999473
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
106.8456418478821
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[2].real | Value:
-135.7143107704046
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[2].imag | Value:
258.7728188264974
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
292.20155015211196
Calculated output path: cases[1].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
117.67492081629787
Calculated output path: cases[1].sourceContributions[1].id | Value: source-B
Calculated output path: cases[1].sourceContributions[1].bus | Value: B
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[0].real | Value:
1845.177379674894
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[0].imag | Value:
-7114.353269196625
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
7349.741628206573
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
-75.4601462566015
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[1].real | Value:
80.07152598510766
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[1].imag | Value:
-264.44816588913363
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
276.3046899994832

Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
-73.15435815211549
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[2].real | Value:
135.7143107703905
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[2].imag | Value:
-258.77281882651005
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
292.2015501521166
Calculated output path: cases[1].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:
-62.32507918370572
Calculated output path: cases[1].kclRelativeResidual | Value: 2.634614910300192e-15
Calculated output path: cases[1].nodalRelativeResidual | Value: 1.1443916996305582e-15
Calculated output path: cases[2].faultBus | Value: F
Calculated output path: cases[2].faultType | Value: BC
Calculated output path: cases[2].maximumPhaseRmsA | Value: 22391.653493161994
Calculated output path: cases[2].groundResidualA | Value: 0
Calculated output path: cases[2].phaseFaultCurrentA[0].real | Value: 0
Calculated output path: cases[2].phaseFaultCurrentA[0].imag | Value: 0
Calculated output path: cases[2].phaseFaultCurrentA[0].magnitude | Value: 0
Calculated output path: cases[2].phaseFaultCurrentA[0].angleDegrees | Value: 0
Calculated output path: cases[2].phaseFaultCurrentA[1].real | Value: -21768.120835010635
Calculated output path: cases[2].phaseFaultCurrentA[1].imag | Value: -5247.386156002774
Calculated output path: cases[2].phaseFaultCurrentA[1].magnitude | Value: 22391.653493161994
Calculated output path: cases[2].phaseFaultCurrentA[1].angleDegrees | Value: -166.44694691812296
Calculated output path: cases[2].phaseFaultCurrentA[2].real | Value: 21768.120835010635
Calculated output path: cases[2].phaseFaultCurrentA[2].imag | Value: 5247.386156002774
Calculated output path: cases[2].phaseFaultCurrentA[2].magnitude | Value: 22391.653493161994
Calculated output path: cases[2].phaseFaultCurrentA[2].angleDegrees | Value: 13.553053081877025
Calculated output path: cases[2].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[2].sourceContributions[0].bus | Value: A
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[0].real | Value:
-4.647940779013652
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[0].imag | Value:
75.2088609544553
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
75.35234647674756
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
93.53640682826376
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[1].real | Value:
-12447.542753653599
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[1].imag | Value:
-3314.8159732921945
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
12881.35573380486
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
-165.0880464235306
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[2].real | Value:
12452.190694432613
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[2].imag | Value:
3239.6071123377387
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
12866.705379887404
Calculated output path: cases[2].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
14.582997428743615
Calculated output path: cases[2].sourceContributions[1].id | Value: source-B
Calculated output path: cases[2].sourceContributions[1].bus | Value: B
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[0].real | Value:
4.647940779007513
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[0].imag | Value:

-75.20886095444166
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
75.35234647673356
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
-86.46359317174026
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[1].real | Value:
-9320.57808135703
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[1].imag | Value:
-1932.5701827105909
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
9518.823629092787
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
-168.2860234845558
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[2].real | Value:
9315.930140578022
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[2].imag | Value:
2007.7790436650323
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
9529.833737915395
Calculated output path: cases[2].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:
12.162412040065265
Calculated output path: cases[2].kclRelativeResidual | Value: 7.043959115381051e-16
Calculated output path: cases[2].nodalRelativeResidual | Value: 1.1443916996305582e-15
Calculated output path: cases[3].faultBus | Value: F
Calculated output path: cases[3].faultType | Value: BCG
Calculated output path: cases[3].maximumPhaseRmsA | Value: 23860.42856802407
Calculated output path: cases[3].groundResidualA | Value: 12962.707171420232
Calculated output path: cases[3].phaseFaultCurrentA[0].real | Value: 4.547473508864641e-13
Calculated output path: cases[3].phaseFaultCurrentA[0].imag | Value: 0
Calculated output path: cases[3].phaseFaultCurrentA[0].magnitude | Value: 4.547473508864641e-13
Calculated output path: cases[3].phaseFaultCurrentA[0].angleDegrees | Value: 0
Calculated output path: cases[3].phaseFaultCurrentA[1].real | Value: -23842.904327254666
Calculated output path: cases[3].phaseFaultCurrentA[1].imag | Value: 914.3110472699864
Calculated output path: cases[3].phaseFaultCurrentA[1].magnitude | Value: 23860.42856802407
Calculated output path: cases[3].phaseFaultCurrentA[1].angleDegrees | Value: 177.80393747920164
Calculated output path: cases[3].phaseFaultCurrentA[2].real | Value: 20122.239724281775
Calculated output path: cases[3].phaseFaultCurrentA[2].imag | Value: 11502.952424371988
Calculated output path: cases[3].phaseFaultCurrentA[2].magnitude | Value: 23178.05958226074
Calculated output path: cases[3].phaseFaultCurrentA[2].angleDegrees | Value: 29.75460336679494
Calculated output path: cases[3].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[3].sourceContributions[0].bus | Value: A
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[0].real | Value:
82.45092885900658
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[0].imag | Value:
-126.83321050097766
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
151.27729160617028
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
-56.973165808843355
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[1].real | Value:
-13616.047377488154
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[1].imag | Value:
0.3018068593009957
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
13616.047380833008
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
179.9987300088797
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[2].real | Value:
11507.439130769942

Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[2].imag | Value:
6611.292552068544
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
13271.410797553222
Calculated output path: cases[3].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
29.878373668669852
Calculated output path: cases[3].sourceContributions[1].id | Value: source-B
Calculated output path: cases[3].sourceContributions[1].bus | Value: B
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[0].real | Value:
-82.4509288590117
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[0].imag | Value:
126.83321050099221
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
151.27729160618523
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
123.02683419115526
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[1].real | Value:
-10226.856949766505
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[1].imag | Value:
914.0092404106776
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
10267.6197807741
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
174.89284903491014
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[2].real | Value:
8614.800593511834
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[2].imag | Value:
4891.659872303444
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
9906.72123218755
Calculated output path: cases[3].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:
29.58879598235543
Calculated output path: cases[3].kclRelativeResidual | Value: 6.518073792588415e-16
Calculated output path: cases[3].nodalRelativeResidual | Value: 1.1443916996305582e-15
Calculated output path: cases[4].faultBus | Value: A
Calculated output path: cases[4].faultType | Value: 3PH
Calculated output path: cases[4].maximumPhaseRmsA | Value: 30498.57599266589
Calculated output path: cases[4].groundResidualA | Value: 0
Calculated output path: cases[4].phaseFaultCurrentA[0].real | Value: 6236.357264374502
Calculated output path: cases[4].phaseFaultCurrentA[0].imag | Value: -29854.16194857092
Calculated output path: cases[4].phaseFaultCurrentA[0].magnitude | Value: 30498.57599266589
Calculated output path: cases[4].phaseFaultCurrentA[0].angleDegrees | Value: -78.20092334818762
Calculated output path: cases[4].phaseFaultCurrentA[1].real | Value: -28972.641288344403
Calculated output path: cases[4].phaseFaultCurrentA[1].imag | Value: 9526.237156261515
Calculated output path: cases[4].phaseFaultCurrentA[1].magnitude | Value: 30498.57599266589
Calculated output path: cases[4].phaseFaultCurrentA[1].angleDegrees | Value: 161.79907665181238
Calculated output path: cases[4].phaseFaultCurrentA[2].real | Value: 22736.284023969904
Calculated output path: cases[4].phaseFaultCurrentA[2].imag | Value: 20327.924792309404
Calculated output path: cases[4].phaseFaultCurrentA[2].magnitude | Value: 30498.57599266589
Calculated output path: cases[4].phaseFaultCurrentA[2].angleDegrees | Value: 41.799076651812385
Calculated output path: cases[4].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[4].sourceContributions[0].bus | Value: A
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[0].real | Value:
3835.5260771547164
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[0].imag | Value:
-20136.511905062256
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
20498.54560673142
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:

-79.21570213243741
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[1].real | Value:
-19356.493891969054
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[1].imag | Value:
6746.592932837471
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
20498.54560673142
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
160.78429786756263
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[2].real | Value:
15520.967814814336
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[2].imag | Value:
13389.918972224785
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
20498.54560673141
Calculated output path: cases[4].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
40.78429786756261
Calculated output path: cases[4].sourceContributions[1].id | Value: source-B
Calculated output path: cases[4].sourceContributions[1].bus | Value: B
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[0].real | Value:
2400.831187219788
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[0].imag | Value:
-9717.650043508667
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
10009.830805644577
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
-76.12247019679201
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[1].real | Value:
-9616.147396375354
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[1].imag | Value:
2779.644223424044
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
10009.830805644579
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
163.87752980320798
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[2].real | Value:
7215.316209155566
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[2].imag | Value:
6938.005820084623
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
10009.830805644577
Calculated output path: cases[4].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:
43.877529803208006
Calculated output path: cases[4].kclRelativeResidual | Value: 1.4910445359672495e-16
Calculated output path: cases[4].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[5].faultBus | Value: A
Calculated output path: cases[5].faultType | Value: AG
Calculated output path: cases[5].maximumPhaseRmsA | Value: 18480.083814940346
Calculated output path: cases[5].groundResidualA | Value: 18480.08381494035
Calculated output path: cases[5].phaseFaultCurrentA[0].real | Value: 4567.958329592716
Calculated output path: cases[5].phaseFaultCurrentA[0].imag | Value: -17906.62599448385
Calculated output path: cases[5].phaseFaultCurrentA[0].magnitude | Value: 18480.083814940346
Calculated output path: cases[5].phaseFaultCurrentA[0].angleDegrees | Value: -75.68913450005512
Calculated output path: cases[5].phaseFaultCurrentA[1].real | Value: 0
Calculated output path: cases[5].phaseFaultCurrentA[1].imag | Value: 0
Calculated output path: cases[5].phaseFaultCurrentA[1].magnitude | Value: 0
Calculated output path: cases[5].phaseFaultCurrentA[1].angleDegrees | Value: 0
Calculated output path: cases[5].phaseFaultCurrentA[2].real | Value: 0
Calculated output path: cases[5].phaseFaultCurrentA[2].imag | Value: -2.2737367544323206e-13

Calculated output path: cases[5].phaseFaultCurrentA[2].magnitude | Value: 2.2737367544323206e-13
Calculated output path: cases[5].phaseFaultCurrentA[2].angleDegrees | Value: -90
Calculated output path: cases[5].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[5].sourceContributions[0].bus | Value: A
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[0].real | Value:
2809.524448975933
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[0].imag | Value:
-12075.418391622417
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
12397.949708001177
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
-76.90231103553285
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[1].real | Value:
-10.603101383412195
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[1].imag | Value:
105.88130106211065
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
106.41088136817697
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
95.7186142527822
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[2].real | Value:
-109.79732753116195
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[2].imag | Value:
90.68244814731565
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
142.4035095598976
Calculated output path: cases[5].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
140.44644038667025
Calculated output path: cases[5].sourceContributions[1].id | Value: source-B
Calculated output path: cases[5].sourceContributions[1].bus | Value: B
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[0].real | Value:
1758.4338806167862
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[0].imag | Value:
-5831.207602861424
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
6090.572372131364
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
-73.21901563989888
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[1].real | Value:
10.603101383426065
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[1].imag | Value:
-105.88130106211474
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
106.41088136818243
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
-84.28138574721058
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[2].real | Value:
109.797327531149
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[2].imag | Value:
-90.68244814732509
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
142.40350955989362
Calculated output path: cases[5].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:
-39.553559613336
Calculated output path: cases[5].kclRelativeResidual | Value: 2.06998798725602e-15
Calculated output path: cases[5].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[6].faultBus | Value: A
Calculated output path: cases[6].faultType | Value: BC
Calculated output path: cases[6].maximumPhaseRmsA | Value: 25016.887310037237

Calculated output path: cases[6].groundResidualA | Value: 0
Calculated output path: cases[6].phaseFaultCurrentA[0].real | Value: 0
Calculated output path: cases[6].phaseFaultCurrentA[0].imag | Value: 0
Calculated output path: cases[6].phaseFaultCurrentA[0].magnitude | Value: 0
Calculated output path: cases[6].phaseFaultCurrentA[0].angleDegrees | Value: 0
Calculated output path: cases[6].phaseFaultCurrentA[1].real | Value: -24471.76335445784
Calculated output path: cases[6].phaseFaultCurrentA[1].imag | Value: -5193.981999055892
Calculated output path: cases[6].phaseFaultCurrentA[1].magnitude | Value: 25016.887310037237
Calculated output path: cases[6].phaseFaultCurrentA[1].angleDegrees | Value: -168.0171420111533
Calculated output path: cases[6].phaseFaultCurrentA[2].real | Value: 24471.76335445784
Calculated output path: cases[6].phaseFaultCurrentA[2].imag | Value: 5193.981999055892
Calculated output path: cases[6].phaseFaultCurrentA[2].magnitude | Value: 25016.887310037237
Calculated output path: cases[6].phaseFaultCurrentA[2].angleDegrees | Value: 11.982857988846684
Calculated output path: cases[6].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[6].sourceContributions[0].bus | Value: A
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[0].real | Value:
-15.103494997513735
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[0].imag | Value:
135.0062930054446
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
135.84849911652276
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
96.38328196237617
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[1].real | Value:
-16616.448667450753
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[1].imag | Value:
-3279.6918498224954
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
16937.022906869377
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
-168.83468965721022
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[2].real | Value:
16631.552162448264
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[2].imag | Value:
3144.6855568170513
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
16926.239233317345
Calculated output path: cases[6].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
10.707055312687396
Calculated output path: cases[6].sourceContributions[1].id | Value: source-B
Calculated output path: cases[6].sourceContributions[1].bus | Value: B
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[0].real | Value:
15.103494997509188
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[0].imag | Value:
-135.0062930054337
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
135.8484991165114
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
-83.61671803762523
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[1].real | Value:
-7855.314687007081
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[1].imag | Value:
-1914.2901492334067
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
8085.201024548567
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
-166.30433167522247
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[2].real | Value:
7840.211192009572

Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[2].imag | Value:
2049.2964422388404
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
8103.611999811252
Calculated output path: cases[6].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:
14.648403657847984
Calculated output path: cases[6].kclRelativeResidual | Value: 5.047357825608596e-16
Calculated output path: cases[6].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[7].faultBus | Value: A
Calculated output path: cases[7].faultType | Value: BCG
Calculated output path: cases[7].maximumPhaseRmsA | Value: 26616.87717648401
Calculated output path: cases[7].groundResidualA | Value: 14301.780065746188
Calculated output path: cases[7].phaseFaultCurrentA[0].real | Value: -9.094947017729282e-13
Calculated output path: cases[7].phaseFaultCurrentA[0].imag | Value: -1.8189894035458565e-12
Calculated output path: cases[7].phaseFaultCurrentA[0].magnitude | Value: 2.033691978340166e-12
Calculated output path: cases[7].phaseFaultCurrentA[0].angleDegrees | Value: -116.56505117707799
Calculated output path: cases[7].phaseFaultCurrentA[1].real | Value: -26565.04678180434
Calculated output path: cases[7].phaseFaultCurrentA[1].imag | Value: 1660.2530254699905
Calculated output path: cases[7].phaseFaultCurrentA[1].magnitude | Value: 26616.87717648401
Calculated output path: cases[7].phaseFaultCurrentA[1].angleDegrees | Value: 176.42379968288552
Calculated output path: cases[7].phaseFaultCurrentA[2].real | Value: 22806.084097112813
Calculated output path: cases[7].phaseFaultCurrentA[2].imag | Value: 12138.70028933153
Calculated output path: cases[7].phaseFaultCurrentA[2].magnitude | Value: 25835.354004905745
Calculated output path: cases[7].phaseFaultCurrentA[2].angleDegrees | Value: 28.024457675473407
Calculated output path: cases[7].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[7].sourceContributions[0].bus | Value: A
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[0].real | Value:
32.5792077872818
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[0].imag | Value:
60.88909594488723
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
69.05712696769658
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
61.85062796375049
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[1].real | Value:
-17881.197354643777
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[1].imag | Value:
1311.7407912821454
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
17929.246574779238
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
175.8043735013706
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[2].real | Value:
15618.994289668688
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[2].imag | Value:
7785.55952929786
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
17451.87439230938
Calculated output path: cases[7].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
26.49475907293966
Calculated output path: cases[7].sourceContributions[1].id | Value: source-B
Calculated output path: cases[7].sourceContributions[1].bus | Value: B
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[0].real | Value:
-32.579207787286805
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[0].imag | Value:
-60.88909594487541
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
69.05712696768852
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:

-118.1493720362578
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[1].real | Value:
-8683.849427160561
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[1].imag | Value:
348.5122341878364
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
8690.840100414654
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
177.70176006014083
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[2].real | Value:
7187.089807444123
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[2].imag | Value:
4353.14076003367
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
8402.624255370087
Calculated output path: cases[7].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:
31.20282561718744
Calculated output path: cases[7].kclRelativeResidual | Value: 5.595445598483621e-16
Calculated output path: cases[7].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[8].faultBus | Value: B
Calculated output path: cases[8].faultType | Value: 3PH
Calculated output path: cases[8].maximumPhaseRmsA | Value: 26978.282592342242
Calculated output path: cases[8].groundResidualA | Value: 0
Calculated output path: cases[8].phaseFaultCurrentA[0].real | Value: 5868.550604575686
Calculated output path: cases[8].phaseFaultCurrentA[0].imag | Value: -26332.258646645005
Calculated output path: cases[8].phaseFaultCurrentA[0].magnitude | Value: 26978.282592342242
Calculated output path: cases[8].phaseFaultCurrentA[0].angleDegrees | Value: -77.43607729990634
Calculated output path: cases[8].phaseFaultCurrentA[1].real | Value: -25738.680229304857
Calculated output path: cases[8].phaseFaultCurrentA[1].imag | Value: 8083.815416365433
Calculated output path: cases[8].phaseFaultCurrentA[1].magnitude | Value: 26978.28259234223
Calculated output path: cases[8].phaseFaultCurrentA[1].angleDegrees | Value: 162.56392270009366
Calculated output path: cases[8].phaseFaultCurrentA[2].real | Value: 19870.129624729172
Calculated output path: cases[8].phaseFaultCurrentA[2].imag | Value: 18248.443230279572
Calculated output path: cases[8].phaseFaultCurrentA[2].magnitude | Value: 26978.28259234224
Calculated output path: cases[8].phaseFaultCurrentA[2].angleDegrees | Value: 42.563922700093656
Calculated output path: cases[8].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[8].sourceContributions[0].bus | Value: A
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[0].real | Value:
3277.689332087544
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[0].imag | Value:
-12200.288069436963
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
12632.90450905596
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
-74.96218409641774
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[1].real | Value:
-12204.604067664386
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[1].imag | Value:
3261.58180741742
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
12632.904509055958
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
165.03781590358227
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[2].real | Value:
8926.914735576844
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[2].imag | Value:
8938.706262019543
Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
12632.90450905596

Calculated output path: cases[8].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
45.037815903582256
Calculated output path: cases[8].sourceContributions[1].id | Value: source-B
Calculated output path: cases[8].sourceContributions[1].bus | Value: B
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[0].real | Value:
2590.8612724881414
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[0].imag | Value:
-14131.97057720804
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
14367.503420161505
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
-79.61114218453038
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[1].real | Value:
-13534.076161640469
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[1].imag | Value:
4822.233608948012
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
14367.503420161507
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
160.3888578154696
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[2].real | Value:
10943.214889152327
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[2].imag | Value:
9309.736968260027
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
14367.503420161507
Calculated output path: cases[8].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:
40.38885781546963
Calculated output path: cases[8].kclRelativeResidual | Value: 1.3899857073977845e-16
Calculated output path: cases[8].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[9].faultBus | Value: B
Calculated output path: cases[9].faultType | Value: AG
Calculated output path: cases[9].maximumPhaseRmsA | Value: 16994.64683705812
Calculated output path: cases[9].groundResidualA | Value: 16994.64683705812
Calculated output path: cases[9].phaseFaultCurrentA[0].real | Value: 4289.866175680656
Calculated output path: cases[9].phaseFaultCurrentA[0].imag | Value: -16444.302031739768
Calculated output path: cases[9].phaseFaultCurrentA[0].magnitude | Value: 16994.64683705812
Calculated output path: cases[9].phaseFaultCurrentA[0].angleDegrees | Value: -75.37896859472934
Calculated output path: cases[9].phaseFaultCurrentA[1].real | Value: -4.547473508864641e-13
Calculated output path: cases[9].phaseFaultCurrentA[1].imag | Value: 0
Calculated output path: cases[9].phaseFaultCurrentA[1].magnitude | Value: 4.547473508864641e-13
Calculated output path: cases[9].phaseFaultCurrentA[1].angleDegrees | Value: 180
Calculated output path: cases[9].phaseFaultCurrentA[2].real | Value: 0
Calculated output path: cases[9].phaseFaultCurrentA[2].imag | Value: 0
Calculated output path: cases[9].phaseFaultCurrentA[2].magnitude | Value: 0
Calculated output path: cases[9].phaseFaultCurrentA[2].angleDegrees | Value: 0
Calculated output path: cases[9].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[9].sourceContributions[0].bus | Value: A
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[0].real | Value:
2225.1268079002793
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[0].imag | Value:
-7097.0628469756475
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
7437.707332582976
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
-72.5923394137464
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[1].real | Value:
-116.57535786387461
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[1].imag | Value:

470.2199151564748
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
484.45493358087793
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
103.92384429594352
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[2].real | Value:
-81.52622240683968
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[2].imag | Value:
487.6628559840011
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
494.43056746766985
Calculated output path: cases[9].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
99.49079343179898
Calculated output path: cases[9].sourceContributions[1].id | Value: source-B
Calculated output path: cases[9].sourceContributions[1].bus | Value: B
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[0].real | Value:
2064.7393677803784
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[0].imag | Value:
-9347.239184764108
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
9572.56648104633
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
-77.54378953733182
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[1].real | Value:
116.57535786389053
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[1].imag | Value:
-470.2199151564796
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
484.4549335808864
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
-76.07615570405478
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[2].real | Value:
81.52622240682467
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[2].imag | Value:
-487.66285598401225
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
494.43056746767843
Calculated output path: cases[9].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:
-80.50920656820296
Calculated output path: cases[9].kclRelativeResidual | Value: 2.5841133532992237e-15
Calculated output path: cases[9].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[10].faultBus | Value: B
Calculated output path: cases[10].faultType | Value: BC
Calculated output path: cases[10].maximumPhaseRmsA | Value: 21969.261655778915
Calculated output path: cases[10].groundResidualA | Value: 0
Calculated output path: cases[10].phaseFaultCurrentA[0].real | Value: 0
Calculated output path: cases[10].phaseFaultCurrentA[0].imag | Value: 0
Calculated output path: cases[10].phaseFaultCurrentA[0].magnitude | Value: 0
Calculated output path: cases[10].phaseFaultCurrentA[0].angleDegrees | Value: 0
Calculated output path: cases[10].phaseFaultCurrentA[1].real | Value: -21421.371994083645
Calculated output path: cases[10].phaseFaultCurrentA[1].imag | Value: -4875.785023067225
Calculated output path: cases[10].phaseFaultCurrentA[1].magnitude | Value: 21969.261655778915
Calculated output path: cases[10].phaseFaultCurrentA[1].angleDegrees | Value: -167.17719088424568
Calculated output path: cases[10].phaseFaultCurrentA[2].real | Value: 21421.371994083645
Calculated output path: cases[10].phaseFaultCurrentA[2].imag | Value: 4875.785023067225
Calculated output path: cases[10].phaseFaultCurrentA[2].magnitude | Value: 21969.261655778915
Calculated output path: cases[10].phaseFaultCurrentA[2].angleDegrees | Value: 12.822809115754323
Calculated output path: cases[10].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[10].sourceContributions[0].bus | Value: A

Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[0].real | Value:
21.11590185682462
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[0].imag | Value:
-45.99386041049638
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
50.60945076452891
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
-65.34002070720733
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[1].real | Value:
-9893.641715076477
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[1].imag | Value:
-2672.704841921048
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
10248.292421585638
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
-164.88275192611255
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[2].real | Value:
9872.525813219652
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[2].imag | Value:
2718.698702331544
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
10240.023855770432
Calculated output path: cases[10].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
15.396504628016435
Calculated output path: cases[10].sourceContributions[1].id | Value: source-B
Calculated output path: cases[10].sourceContributions[1].bus | Value: B
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[0].real | Value:
-21.115901856830305
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[0].imag | Value:
45.99386041051184
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
50.609450764545336
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
114.6599792927912
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[1].real | Value:
-11527.730279007166
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[1].imag | Value:
-2203.080181146188
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
11736.35921698453
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
-169.1805879849768
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[2].real | Value:
11548.846180863995
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[2].imag | Value:
2157.086320735676
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
11748.568827919504
Calculated output path: cases[10].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value:
10.57976380003714
Calculated output path: cases[10].kclRelativeResidual | Value: 8.183408020502075e-16
Calculated output path: cases[10].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: cases[11].faultBus | Value: B
Calculated output path: cases[11].faultType | Value: BCG
Calculated output path: cases[11].maximumPhaseRmsA | Value: 23492.242200550376
Calculated output path: cases[11].groundResidualA | Value: 13543.20300812607
Calculated output path: cases[11].phaseFaultCurrentA[0].real | Value: -4.547473508864641e-13
Calculated output path: cases[11].phaseFaultCurrentA[0].imag | Value: 0
Calculated output path: cases[11].phaseFaultCurrentA[0].magnitude | Value: 4.547473508864641e-13

Calculated output path: cases[11].phaseFaultCurrentA[0].angleDegrees | Value: 180
Calculated output path: cases[11].phaseFaultCurrentA[1].real | Value: -23436.97494498932
Calculated output path: cases[11].phaseFaultCurrentA[1].imag | Value: 1610.4809956229947
Calculated output path: cases[11].phaseFaultCurrentA[1].magnitude | Value: 23492.242200550376
Calculated output path: cases[11].phaseFaultCurrentA[1].angleDegrees | Value: 176.06907711415565
Calculated output path: cases[11].phaseFaultCurrentA[2].real | Value: 19864.811207753264
Calculated output path: cases[11].phaseFaultCurrentA[2].imag | Value: 11453.13236185551
Calculated output path: cases[11].phaseFaultCurrentA[2].magnitude | Value: 22930.00142210771
Calculated output path: cases[11].phaseFaultCurrentA[2].angleDegrees | Value: 29.965762397037654
Calculated output path: cases[11].sourceContributions[0].id | Value: grid-A
Calculated output path: cases[11].sourceContributions[0].bus | Value: A
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[0].real | Value:
104.73449676442351
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[0].imag | Value:
-427.21053738685805
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[0].magnitude | Value:
439.86152146654644
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[0].angleDegrees | Value:
-76.2251069047261
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[1].real | Value:
-10884.743151181925
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[1].imag | Value:
-74.87305833482606
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[1].magnitude | Value:
10885.000663393008
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[1].angleDegrees | Value:
-179.6058848130228
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[2].real | Value:
9107.359035811514
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[2].imag | Value:
5375.166788793528
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[2].magnitude | Value:
10575.273358855862
Calculated output path: cases[11].sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value:
30.549120597101183
Calculated output path: cases[11].sourceContributions[1].id | Value: source-B
Calculated output path: cases[11].sourceContributions[1].bus | Value: B
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[0].real | Value:
-104.7344967644301
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[0].imag | Value:
427.2105373868735
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[0].magnitude | Value:
439.86152146656303
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value:
103.77489309527427
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[1].real | Value:
-12552.23179380739
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[1].imag | Value:
1685.354053957813
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[1].magnitude | Value:
12664.870362252475
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value:
172.3527864210715
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[2].real | Value:
10757.45217194175
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[2].imag | Value:
6077.96557306198
Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[2].magnitude | Value:
12355.745333201916

Calculated output path: cases[11].sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: 29.466468849010106
Calculated output path: cases[11].kclRelativeResidual | Value: 6.12732485453742e-16
Calculated output path: cases[11].nodalRelativeResidual | Value: 1.0546329814172563e-15
Calculated output path: limitations[0] | Value: Only the listed fault locations and types are evaluated
Calculated output path: limitations[1] | Value: All cases use identical documented network and source operating conditions
Calculated output path: limitations[2] | Value: No IEC automatic maximum/minimum, voltage correction, transformer vector group, machine transient or breaker duty
Calculated output path: limitations[3] | Value: Any undefined zero-sequence ground-return path causes entire sweep to fail closed

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: faultsweep
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:34.981Z
Evidence / limitation: outputs.scope | Value: Fault-location and fault-type grid over one explicitly entered fixed source operating condition. NOT automatic IEC 60909 maximum/minimum source conditions, nor breaker duty or protection certification.
Evidence / limitation: outputs.limitations[0] | Value: Only the listed fault locations and types are evaluated
Evidence / limitation: outputs.limitations[1] | Value: All cases use identical documented network and source operating conditions
Evidence / limitation: outputs.limitations[2] | Value: No IEC automatic maximum/minimum, voltage correction, transformer vector group, machine transient or breaker duty
Evidence / limitation: outputs.limitations[3] | Value: Any undefined zero-sequence ground-return path causes entire sweep to fail closed